LOGESHWARAN C

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CAREER OBJECTIVE

Seeking an entry-level opportunity in Electrical and Electronics Engineering, Embedded Systems, or Firmware Engineering to apply my knowledge of electrical and electronics engineering, Electric Vehicle (EV) technology, embedded systems, Embedded C, and microcontrollers. Eager to contribute to real-world projects, develop practical expertise, and grow as a versatile engineering professional.

EDUCATION

B.E., (Hons.,) Electrical and Electronics Engineering — University College of Engineering Panruti, Anna University
CGPA: 8.31 | 2021–2025
Class XII – St.Paul's Matriculation Higher secondary school, Neyveli
Score : 86.1% | 2020–2021
Class X - Tripuraneni Vidyalaya Matric School, Vadalur
Score : 81.4% | 2018–2019

TECHNICAL SKILLS

- **Programming:** Advanced C, C++ (OOP), Data Structures
- **Systems:** Linux system calls, IPC (pipes, FIFO, shared memory), Linux command line
- **Tools:** Vim, Makefiles, GCC, XC8, Git
- **Embedded and Hardware:** Arduino, LoRa, sensor interfacing, microcontrollers, EV architecture (BMS, controllers)

TRAINING AND CERTIFICATIONS

- **Advanced Embedded Systems Course** — Emertxe Information Technologies, Bengaluru (Government of India / NSDC certified, Embedded Software Engineer)
- **Electric Vehicle Technology Training** — MobilityNxt India Pvt. Ltd., under Naan Mudhalvan Sigram Thodu **(Government of Tamil Nadu)**

PROJECTS

Autonomous Robotic Boat for Marine Waste Collection *Tools:* **Arduino, LiDAR, Sensors, Embedded C**

- Built an autonomous and Bluetooth-controlled robotic boat with LiDAR-based obstacle detection and metal/moisture sensors for marine waste identification.
- Integrated LiDAR and sensor inputs with Arduino to enable real-time obstacle detection and waste identification.
- Ranked in the Top 50 among 8,000+ teams in the Naan Mudhalvan Innovation Challenge.
- Received a ₹10,000 government project grant for the project.

Smart Bus Route Display System *Tools:* **Arduino, LoRa, IR Sensors, C , wireless**

- Developed a real-time bus occupancy and route display system using LoRa communication and IR-based passenger counting.
- Designed TX / RX units to exchange route name, distance, and seat-availability data over long-range LoRa links.
- Resolved transmitter-receiver packet synchronization issues to ensure accurate and timely occupancy updates.
- Calibrated IR sensors at entry and exit points to accurately track passenger counts in real time.

Image Steganography Using LSB *Tools:* **C, Bit Manipulation, BMP File Handling**

- Developed an image steganography system to hide and extract text using least-significant-bit manipulation with minimal visual distortion.
- Parsed raw BMP file structures and header offsets at the byte level to locate and process pixel data.
- Applied bit masking and bitwise operations to embed data without corrupting or visibly altering the image.
- Verified extraction accuracy by decoding hidden text and confirming zero data loss.

Arbitrary Precision Calculator (APC) *Tools:* **Advanced C, Doubly Linked Lists**

- Implemented arbitrary-precision integer arithmetic using doubly linked lists to support numbers beyond native data type.
- Implemented sign management and result normalization, including leading-zero removal, for consistent output formatting.
- Managed dynamic memory and prevented dangling pointers.
- Supported addition, subtraction, and multiplication of large integers exceeding standard data type limits.

INTERNSHIPS

NLC India Limited — Thermal Power Station-I Expansion (TPS-I EXP), Neyveli

- Studied the 440 kV switchyard and associated high-voltage electrical equipment.
- Gained practical exposure to switchyard equipment, electrical systems, and their operation.

NLC India Limited — New Service Unit (NSU), Neyveli

- Studied transformers and motors, including their maintenance, repair, and overhauling procedures.
- Gained practical exposure to electrical equipment servicing and industrial maintenance practices.

ACHIEVEMENTS

• 1st Place — Paper Presentation, EGS Pillay Engineering College, Nagapattinam
• 1st Place — Quiz Competition, Institution of Engineers (India), Neyveli Local Centre
• 3rd Place — Paper Presentation, Anna University, Tindivanam

LANGUAGES

- English – Proficient
- Tamil – Proficient